between entangled, unentangled, and partially entangled states by creating "rap sheets" for three specific examples—the singlet state, a product state, and a "near singlet" state. We hope this format will help clarify the mathematical similarities and differences. Please take some time to review this material and work the exercises before moving on.

State-Vector Rap Sheet 1

Name: Product State (No Entanglement)

Wanted for: Excessive Locality, Impersonating a Classical System

Description: Each subsystem is fully characterized. There are no correlations between Alice's and Bob's systems.

State-Vector: $\alpha_u\beta_u|uu> + \alpha_u\beta_d|ud> + \alpha_d\beta_u|du> + \alpha_d\beta_d|dd>$

Normalization: $\alpha_u^*\alpha_u + \alpha_d^*\alpha_d = 1, \quad \beta_u^*\beta_u + \beta_d^*\beta_d = 1$

Density Matrix: Alice's density matrix has exactly one nonzero eigenvalue, which equals 1. The eigenvector with this nonzero eigenvalue is the wave function of Alice's subsystem. The same goes for Bob.

Wave Function: Factorized: $\psi(a)\varphi(b)$

Expectation Values:

$$\langle\sigma_x\rangle^2 + \langle\sigma_y\rangle^2 + \langle\sigma_z\rangle^2 = 1$$
$$\langle\tau_x\rangle^2 + \langle\tau_y\rangle^2 + \langle\tau_z\rangle^2 = 1$$

Correlation: $\langle\sigma_z\tau_z\rangle - \langle\sigma_z\rangle\langle\tau_z\rangle = 0$